

Md. Azim Sikder

Electrical and Electronic Engineering Graduate
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Research Interests

Digital VLSI and RTL Design, Digital Logic Architecture, Design Verification, FPGA-Based Hardware Design, and Hardware Security, including quantum-secure authentication.

Education

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Electrical and Electronic Engineering

CGPA: 3.89/4.00

June 2026

VLSI and Digital Design Projects

- **Adaptive SPI Communication Interface (Full VLSI Flow)** *Cadence EDA, SystemVerilog*
Took an SPI master/slave interface through the complete digital design flow: RTL design, functional verification, logic synthesis, and physical layout, achieving timing closure and DRC/LVS compliance.
- **Microcontroller-Free Smart Toll Collection System** *Digital Logic Design*
Designed a complete toll collection system implemented entirely in digital logic, without a microcontroller. Submitted to ICECE 2026 [1].
- **Class-A Common-Emitter RF Power Amplifier (433 MHz ISM Band)** *Cadence Virtuoso*
Designed biasing and matching networks for stability ($K > 1$); characterized S-parameters, P1dB, and PAE.

Other Projects:

- **Embedded Systems:** Smart alert system for natural hazards with real-time P-wave detection for seismic early warning; MopBot autonomous cleaning robot.
- **Power Electronics:** Arduino-controlled MPPT seven-level inverter for solar power generation [2].

Industry Training

VLSI Industry Attachment Program, Siliconova

November 2025

- Hands-on training across the chip design flow: RTL design, Verilog HDL, SystemVerilog, verification, DFT, ATPG, physical design, analog layout, DRC/LVS, and IC packaging in the Cadence environment.

Conference Papers (Under Review)

- [1] **M. A. Sikder et al.**, “A Microcontroller-Free Digital Logic Architecture for Smart Toll Collection Systems,” submitted to *2026 14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh.
- [2] **M. A. Sikder et al.**, “An Arduino-Controlled MPPT Seven-Level Inverter for Solar Power Generation,” submitted to *2026 14th International Conference on Electrical and Computer Engineering (ICECE)*, Dhaka, Bangladesh.

Undergraduate Thesis

Quantum Token for Anti-Counterfeit Systems

April 2025 – June 2026

(Under the supervision of Dr. Sajid Muhaimin Choudhury)

- Designed a hybrid quantum-classical authentication scheme combining BB84 encoding with Learning With Errors (LWE), achieving a **100% valid user** and **0% adversary success rate** at **1,500 qubits**, versus the $\sim 4.2 \times 10^6$ qubits required by pure quantum tokens.
- Benchmarked on the **156-qubit IBM Fez** (Heron r2) processor via Qiskit Runtime, with an adaptive verification algorithm that separates hardware noise from tampering.

Technical Skills

HDL: Verilog HDL, SystemVerilog

Programming: C/C++, Python, MATLAB, ARM Assembly, RISC-V Assembly, Qiskit

EDA and Simulation Tools: Cadence (Virtuoso and digital implementation flow), Quartus, PSpice, Proteus, MATLAB/Simulink

Hardware Platforms: FPGA, STM32, Arduino, ESP32

Teaching and Mentoring Experience

Independent Private Tutor

2022 – Present

- Taught undergraduate Electrical Engineering, Mathematics, and Physics to students from Stony Brook University (USA) and the University of Auckland (New Zealand).
- Supported students with coursework, assignments, and exam preparation.

Scholarships and Awards

- Dean's List Scholarship, BUET
- University Merit Stipend Scholarship, BUET
- Champion, Integration Bee Competition organized by Tachyon (2022)
- Exceptional Mentorship Award, Brain Stormers Academic and Admission Care (2023–2024)

Leadership and Activities

- Organizer and Problem Setter, Integration Bee, EEE Day 2026, BUET
- Member, BUET Robotics Club

References

Dr. Md. Ziaur Rahman Khan

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Dr. Sajid Muhaimin Choudhury

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